

SM Graded Quiz M7

Question 1

According to Lesson 1, what type of graph has no self-loops, no parallel edges, and symmetric relationships?

- ☐ Complete graph
- ☐ Directed graph
- ☐ Weighted graph
- ☒ Simple graph

Question 2

In weighted graphs, what do the non-zero entries of the adjacency matrix represent instead of binary values?

- ☐ The weight or strength of each edge
- ☐ The number of parallel edges
- ☐ The direction of the edge
- ☒ The node degree

Question 3

An undirected graph has 10 nodes and 15 edges. What is its density D ?

- ☐ 0.333
- ☐ 0.15
- ☐ 1,500
- ☒ 0.167

Question 4

Given adjacency matrix

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \text{ how}$$

many edges does this undirected graph have?

- ☐ 2
- ☐ 3
- ☐ 1
- ☒ 4

Question 5

What does "out-strength"
 $s^{out}(v) = \sum_u w_{vu}$
measure in a directed
weighted interbank
network?

- ☐ Number of banks
that lend to bank v
- ☐ Total amount of
loans made by bank
 v
- ☐ Total amount
borrowed by bank v
- ☒ Average loan size
for bank v

Question 6

In the ER model $G(n, p)$,
what distribution does
the degree follow for
large n and small p ?

- ☐ Poisson distribution
- ☐ Exponential
distribution
- ☐ Uniform distribution
- ☒ Power-law
distribution

Question 7

Why does Lesson 2 state
that the clustering
coefficient in ER
networks equals the
connection probability p ?

- ☐ Each node connects to a fraction p of all other nodes
- ☐ Each pair of a node's neighbors connects independently with probability p
- ☐ The network has exactly $p \cdot n$ triangles
- ☒ All nodes have exactly p neighbors

Question 8

If the shortest path length between two banks in a similarity network is 2.17 (using inverse cosine similarity), what does this indicate?

- ☐ The banks have no common borrowers
- ☐ The banks are in different connected components
- ☐ The banks are exactly 2 intermediaries apart
- ☒ The banks have moderate similarity in their lending portfolios



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QUESTIONS



Question 9

What is the average path length scaling in BA networks?

☐ $L \sim \ln^2 n$

☐ $L \sim n$

☐ $L \sim \frac{\ln n}{\ln \ln n}$

☒ $L \sim \sqrt{n}$

Question 10

What real-world financial example demonstrates preferential attachment according to Lesson 3?

- ☐ Interest rates are set by central banks
- ☐ Large banks attract more counterparties
- ☐ Markets close on weekends
- ☒ Stock prices follow random walks

Question 11

How does the degree distribution of BA networks differ from ER networks on a log-log plot?

- ☐ Both show bell-shaped curves
- ☐ BA shows a straight line while ER shows rapid exponential cutoff
- ☐ Both show identical straight lines

- ☒ ER shows a straight line while BA curves down

Question 12

If the degree distribution follows $P(k) = 2m^2k^{-3}$ with $m = 3$, what is $P(k = 9)$?

- ☐ $P(9) = \frac{1}{81}$
- ☐ $P(9) = \frac{2}{27}$
- ☐ $P(9) = \frac{18}{729}$
- ☒ $P(9) = \frac{1}{9}$

Question 13

What is the relationship between adjacency matrix powers and path counts?

- ☐ $(A^k)_{ij}$ gives the number of paths of length k from node i to node j
- ☐ $(A^k)_{ij}$ gives the shortest path from i to j
- ☐ $(A^k)_{ij}$ gives the clustering coefficient
- ☒ $(A^k)_{ij}$ gives the degree of node i

Question 14

What power-law exponent does the BA model predict for degree distribution?

- ☐ $P(k) \propto k^{-1}$
- ☐ $P(k) \propto k^{-3}$
- ☐ $P(k) \propto k^{-2}$
- ☒ $P(k) \propto k^{-4}$

Question 15

Based on Lesson 4's findings, should regulators focus more on institution size or network centrality for systemic risk?

- ☐ Network centrality, as it captures interconnectedness beyond just size
- ☐ Focus on profitability instead
- ☐ Neither, as systemic risk is random
- ☒ Size only, as large banks are always systemically important

Question 16

Why does correlation contagion during crises reduce portfolio diversification benefits?

- ☐ The number of available assets decreases
- ☐ Individual stock volatility decreases
- ☐ Assets move together more strongly, eliminating independent movements
- ☒ Trading volumes become too low

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